

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Introduction to Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Ratios of standard angles

1. If $\cos(\alpha + \beta) = 0$, then the value of $\cos\left(\frac{\alpha + \beta}{2}\right)$ is equal to: (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$

CBSE 2024, 1 mark

SOLUTION

 $\cos(\alpha + \beta) = 0 = \cos 90^\circ$, so $\alpha + \beta = 90^\circ$.

$$\cos\left(\frac{\alpha + \beta}{2}\right) = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Answer: (a)

2. Evaluate: $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$ CBSE 2024, 2 marks

SOLUTION

$$\begin{aligned} &= 2\sqrt{2} \times \frac{1}{\sqrt{2}} \times \frac{1}{2} + 2\sqrt{3} \times \frac{\sqrt{3}}{2} \\ &= 1 + 3 \end{aligned}$$

Answer: 4

3. If $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A + B) = \sin A \cos B + \cos A \sin B$. CBSE 2024, 2 marks

SOLUTION

$$\text{LHS} = \sin(60^\circ + 30^\circ) = \sin 90^\circ = 1$$

$$\text{RHS} = \sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1$$

LHS = RHS. Verified.

2 Trigonometric identities

4. If $\sec \theta - \tan \theta = m$, then the value of $\sec \theta + \tan \theta$ is: (a) $1 - \frac{1}{m}$ (b) $m^2 - 1$ (c) $\frac{1}{m}$ (d) $-m$

CBSE 2024, 1 mark

SOLUTION

$\sec^2 \theta - \tan^2 \theta = 1$, so $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$.

$$\sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta} = \frac{1}{m}$$

Answer: (c)

5. Prove that: $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$ **CBSE 2024, 3 marks**

SOLUTION

Write $s = \sin \theta$ and $c = \cos \theta$. Then $\frac{\tan \theta}{1 - \cot \theta} = \frac{s/c}{(s-c)/s} = \frac{s^2}{c(s-c)}$ and $\frac{\cot \theta}{1 - \tan \theta} = \frac{c/s}{(c-s)/c} = \frac{c^2}{s(s-c)}$.

$$\text{LHS} = \frac{s^3 - c^3}{sc(s-c)} = \frac{(s-c)(s^2 + sc + c^2)}{sc(s-c)} = \frac{s^2 + c^2 + sc}{sc}$$

$$= \frac{1 + sc}{sc} = \frac{1}{sc} + 1$$

$$= 1 + \sec \theta \operatorname{cosec} \theta = \text{RHS}$$